

Muhammed Muminul Hoque

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House no. 549/I, Moushumi R/A, DC Road, West Bakalia, Chattogram, Bangladesh

Education

Bachelor of Science in Computer Science & Engineering

University of Dhaka, Bangladesh

February 2019 – March 2024

GPA: 3.35/4.00
(3.58 in last 60 credits)

Research Experience

Research Intern

November 2025 – Present

Under the supervision of *Professor Yanfu Zhang*

Department of Computer Science, William & Mary, Williamsburg, VA

- Developing a "Position-Aware" In-Context Learning (ICL) evaluation pipeline for long-context LLMs (Qwen2.5-7B) to mitigate performance degradation ("Context Rot") in multi-shot prompting.
- Engineering automated, large-scale inference pipelines via **vLLM** on NVIDIA RTX A6000 (48GB VRAM) to test whether ordering candidate examples using the "Lost in the Middle" attention bias yields zero-cost accuracy gains.
- Co-authored *DyV2X*, a dynamic graph learning framework for V2X cooperative trajectory prediction, submitted to IROS 2026 (**2nd author**).
- Designed a GRU history encoder across 6 CTDG architectures (TGN, TGAT, GraphMixer, DyGFormer, DyRep, JODIE) on the V2X-Seq-TFD dataset; achieved minADE = 0.952 m, minFDE = 1.664 m, MR = 24.7%.

Publications

Published

E. Hossain, P. R. Saha, **M. M. Hoque**, M. S. Islam, J. Al Faruk, and M. A. Hossain, 2025
“*Privacy Preserving and Secure Digital Identity Verification Mechanism: AES-Encrypted Smart NID with Two-Factor Protection Against Unauthorized Access,*”
in *Proc. 4th Int. Conf. on Electrical, Computer and Communication Engineering (ECCE)*, CUET, Chattogram, Bangladesh, 2025, pp. 1–6.
doi:10.1109/ECCE64574.2025.11012996

P. R. Saha, Md. S. Islam, T. K. Saha, **M. M. Hoque**, A. Haque, E. Hossain, 2025
“*Quantum-Resistant Blockchain: Enhancing Blockchain Security with ROUND5, A Lattice-based Cryptosystem,*”
in *Proc. IEEE 7th Int. Conf. on Sustainable Technologies for Industry 5.0 (STI 2025)*.
doi:10.1109/STI69347.2025.11367567

- Implemented ROUND5 lattice-based cryptography to reduce post-quantum vulnerability; proposed an efficient key exchange protocol suitable for SMEs.

Accepted

Jonayed Al Faruk, **Muhammed Muminul Hoque**, Md. Nawaz Shorif, Susmoy Paul, 2026
Md. Bodrul Islam, Joyonto Das,
“*Toward Accurate Urban Temperature Forecasting in Seoul: A Hybrid Ensemble of Bias-Corrected Machine and Deep Learning Models with Explainable AI,*”
in *Proc. 3rd Int. Conf. on Big Data, IoT and Machine Learning (BIM 2025)*, Springer Lecture Notes in Networks and Systems, Vol. 1, 2026, accepted for publication.

- Explored XGBoost, KNN, CNN, SVR, and Feedforward Neural Networks; applied SHAP and LIME for explainability of meteorological and topographic features.

Muhammed Muminul Hoque, Md. Masudur Rahman, Jonayed Al Faruk, A.K.M. Bahalul Haque, Mir Tasrif Ahmed, MD. Nayeem Pervez 2026
“A Comparative Analysis of Traditional and Ensemble Machine Learning Models for Phishing Email Detection with Explainable AI.”
 in *Proc. Int. Conf. on Electrical, Computer and Communication Technologies (ECCT 2026)*, Dhaka International University, Bangladesh, May 7–9, 2026. To be published in a Taylor & Francis Book.

- Built a unified dataset of 290,404 labeled emails from 12 public corpora.
- Benchmarked classical vs. ensemble ML models with explainable AI (XAI); achieved ROC-AUC of 94.6%.

Submitted & Under Review

Yiran Ding, **Muhammed Muminul Hoque**, Maiqi Jiang, Yongming Qin, Sidi Lu, [Submitted]
 Yanfu Zhang

“DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction”
 Submitted to *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)*, Pittsburgh, PA, Sept 27–Oct 1, 2026.

- Proposed a dynamic interaction graph framework that captures the temporal evolution of inter-agent interactions for V2X cooperative motion forecasting.
- Demonstrated consistent outperformance over prior methods, with particularly significant gains in safety-critical turning scenarios.

Selected Research in Progress

M. M. Hoque, Md. Masudur Rahman, A.K.M. Bahalul Haque. **“ZEAL: A Zero-Shot Heterogeneous Ensemble Framework for Adaptive Anomaly Detection.”** *Target: IEEE T-IFS or similar Q1 venue.* [In Prep]

- Developed a multi-tier agentic pipeline utilizing heterogeneous model ensembles and reasoning escalation for zero-shot detection — zero training data required.
- Achieves 98.44% accuracy on 4,812 samples across two benchmarks, outperforming fine-tuned transformer baselines through advanced logical consensus.

M. M. Hoque, Maiqi Jiang, Y. Zhang, et al., **“Optimizing In-Context Learning via Non-Linear Attentional Priors in Long-Context Windows.”** *Target: ACL 2026 or NAACL 2026.* [In Prep]

- Proposed an inference-only method using empirical attentional analysis to optimize example positioning in long-context ICL, significantly recovering performance lost in mid-context.
- Benchmarked on state-of-the-art LLMs (7B+) across 4k+ token context lengths, demonstrating consistent accuracy gains over naive relevance-based ordering.

Project Work

Aurora: Autonomous Academic Research Agent optimized for low-resource servers 2026

- Engineered a multi-agent AI system optimized for heavily constrained edge hardware (1GB RAM), utilizing Open-Router to dynamically route tasks between specialized LLMs (Hermes 405B, Qwen, Gemini Flash) with zero local memory overhead.
- Built an asynchronous Telegram interface featuring real-time token streaming, computer vision (document parsing), and automated synthesis of ArXiv research papers.
- Implemented a robust background task scheduler integrating multi-backend deadline tracking across local JSON, Notion APIs, and Google Calendar.
- Tools: Python, AsyncIO, python-telegram-bot, APScheduler, Gemini API.

PeerTranslate: Verified Research Paper Translation for Every Language 2026

- Open-source AI translation tool for research papers with a 4-pass back-translation verification pipeline (Translate → Back-Translate → Score → Re-translate), producing per-section confidence scores.

- Powered by Google Gemini with community-curated domain glossaries (ML, CS, general academic) to prevent mistranslation of technical jargon; supports multiple languages with open contributions.
- Tools: Python, FastAPI, Google Gemini API.
- Live demo: peertranslate.onrender.com

Toxicity Analysis of 4chan's /pol/ Board Using OpenAI Moderation and Google Perspective APIs 2025

- Developed an end-to-end pipeline to detect and analyze toxic speech patterns using OpenAI Moderation API and Google Perspective API for multi-dimensional toxicity scoring.
- Conducted ethical data collection, quantitative analysis, and visualization to identify trends in online discourse.
- Tools: Python, Pandas, Matplotlib.
- *Reviewed by Prof. Kai-Cheng Yang (Binghamton University).*

Machine Learning Based Speech Emotion Recognizer 2024

- Identifies and classifies emotions from speech, providing real-time emotional insights.
- Tools: JupyterLab, Python.
- Supervisor: Prof. Md. Alamgir Hossain, Professor (CSE) & Principal.

Technical Skills

Languages & Frameworks: Python, C/C++, Bash, PyTorch

Graph Neural Networks & Motion Forecasting: TGN, TGAT, GraphMixer, DyGFormer, GCN, GAT, PyG, DGL, HiVT, DenseTNT, V2X-Graph, WOMD evaluation

ML & Explainability: Scikit-learn, SHAP, LIME, GNNExplainer, XGBoost, Random Forest, SVM

Tools & Libraries: Pandas, NumPy, Matplotlib, Git, GitHub, Overleaf, Google Colab

Training Experiences & Certifications

Cyber Security For Professionals, Byte Capsule Feb 2023 – Aug 2023

- Six-month program covering threat analysis, risk management, network security, cryptography, and ethical hacking through hands-on exercises and expert-led lectures.

Standardized Test Scores

International English Language Testing System (IELTS) 14 December 2024

- Overall: 6.5 Listening: 8.0 Reading: 6.0 Writing: 6.0 Speaking: 6.0

Extra-curricular Activities, Awards, and Achievements

- MEC Computer Club (January 2023 – March 2024)
- 3rd Place at MEC CSE Carnival 2020 Programming Competition
- Batch Topper – 6-Month Self-learning Cybersecurity Training Program (Feb – Aug 2023)
- Champion of Invisible CTF 2023 (18 September 2023)
- Ranked in the Top 6% on TryHackMe
- Solved 95+ challenges in PicoCTF, including 50+ medium-level problems across cryptography, reverse engineering, and web security

References

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